Interpretation of IQR and Outlier Boundaries

Academic Percentages and Salary

This assignment interprets the latest statistical table. The sl_no column is excluded because it is only a serial identifier.

Data set file name: Placement.csv

1. Meaning of Statistics

- **IQR:** Interquartile range, calculated as Q3 − Q1. It measures the spread of the middle 50% of observations.
- **1.5 × IQR:** The interquartile range multiplied by 1.5; it is used to calculate potential-outlier boundaries.
- **Lesser Outliers:** Q1 − (1.5 × IQR). Observations below this boundary are potential lower outliers.
- **Greater Outliers:** Q3 + (1.5 × IQR). Observations above this boundary are potential upper outliers.
- **Minimum / Maximum:** The smallest and largest observed values in each variable.

2. Values in the Table

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
IQR	107.0	15.1	12.1	11.0	23.5	8.31	60000.0
1.5 Rule	160.5	22.65	18.15	16.5	35.25	12.465	90000.0
Lesser	-106.0	37.95	42.75	44.5	24.75	45.48	150000.0
Greater	322.0	98.35	91.15	88.5	118.75	78.72	390000.0
Min	1	40.89	37.0	50.0	50.0	51.21	200000.0
Max	215	89.4	97.7	91.0	98.0	77.89	940000.0

3. How to Calculate the Boundaries

1. Step 1: IQR = Q3 − Q1.
2. Step 2: Calculate 1.5 × IQR.
3. Step 3: Lesser Outliers = Q1 − (1.5 × IQR).
4. Step 4: Greater Outliers = Q3 + (1.5 × IQR).
5. Step 5: Compare the observed minimum and maximum with the boundaries. Values below the lesser Outliers or above the Greater Outliers are potential outliers.

4. Interpretation of Each Variable

3.1 SSC Percentage (ssc_p)

- IQR = 15.1 percentage points. The middle 50% spans from Q1 = 60.6% to Q3 = 75.7%.
- 1.5 × IQR = 22.65. Lesser Outlier = 37.95%; Greater Outlier = 98.35%.
- Minimum = 40.89%, which is above the lesser Outlier. Maximum = 89.4%, which is below the Greater Outlier.
- Interpretation: Neither the minimum nor maximum is a potential outlier under the 1.5 × IQR rule.

3.2 HSC Percentage (hsc_p)

- IQR = 12.1 percentage points. The middle 50% spans from Q1 = 60.9% to Q3 = 73%.
- 1.5 × IQR = 18.15. Lesser Outlier = 42.75%; Greater Outlier = 91.15%.
- Minimum = 37%, below the lesser Outlier. Maximum = 97.7%, above the Greater Outlier.
- Interpretation: Both the minimum and maximum are potential outliers under the 1.5 × IQR rule.

3.3 Degree Percentage (degree_p)

- IQR = 11 percentage points. The middle 50% spans from Q1 = 61% to Q3 = 72%.
- 1.5 × IQR = 16.5. Lesser Outlier = 44.5%; Greater Outlier = 88.5%.
- Minimum = 50%, above the lesser Outlier. Maximum = 91%, above the Greater Outlier.
- Interpretation: The maximum is a potential Greater outlier; the minimum is not a potential lesser outlier.

3.4 Employability Test Percentage (etest_p)

- IQR = 23.5 percentage points. The middle 50% spans from Q1 = 60% to Q3 = 83.5%.
- 1.5 × IQR = 35.25. Lesser Outlier = 24.75%; Greater Outlier = 118.75%.
- Minimum = 50% and maximum = 98%; both values are within the boundaries.
- Interpretation: Neither the minimum nor maximum is a potential outlier under this rule.

3.5 MBA Percentage (mba_p)

- IQR = 8.31 percentage points. The middle 50% spans from Q1 = 57.945% to Q3 = 66.255%.
- 1.5 × IQR = 12.465. Lesser Outlier = 45.48%; Greater Outlier = 78.72%.
- Minimum = 51.21% and maximum = 77.89%; both values are within the boundaries.
- Interpretation: Neither the minimum nor maximum is a potential outlier under this rule.

3.6 Salary

- IQR = 60,000 salary units. The middle 50% spans from Q1 = 240,000 to Q3 = 300,000.
- 1.5 × IQR = 90,000. Lesser Outlier = 150,000; Greater Outlier = 390,000.
- Minimum salary = 200,000, above the lesser Outlier. Maximum salary = 940,000, above the Greater Outlier.
- Interpretation: The maximum salary is a potential upper outlier. The currency and salary period are not specified, so they should not be assumed.

5. Key Findings

- Employability-test percentage has the largest IQR among the academic variables (23.5 percentage points), meaning its middle 50% has the widest spread in absolute percentage-point terms.
- MBA percentage has the smallest academic IQR (8.31 percentage points), meaning its middle 50% has the narrowest spread.
- HSC has both a minimum below its lesser Outlier and a maximum above its Greater Outlier.
- Degree percentage has a maximum above its Greater Outlier.
- Salary has a maximum of 940,000, which is well above its Greater Outlier of 390,000.

- SSC, employability-test, and MBA minimum and maximum values are within their respective boundaries.
- A potential outlier is not automatically an error. It should be investigated against the original data before any decision is made.

6. Conclusion

The IQR describes the spread of the middle 50% of observations. The 1.5 × IQR rule creates lesser and Greater Outliers for identifying potential outliers. In this table, both HSC extremes, the degree maximum, and the salary maximum fall outside their relevant boundaries. These values should be checked, not automatically removed. The other listed minimum and maximum values fall within their boundaries.